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Review

Smart manufacturing powered by recent technological advancements: A review

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ABSTRACT

Smart manufacturing has attracted significant attention from both researchers and manufacturing experts owing to its capability of accomplish the goals of Industry 4.0. It significantly contributes to build highly effective smart factories, increase yield, cut down human error by utilizing autonomous machines or robots, optimize overall electrical consumption, fulfill users' customized demands, etc. Hence, we reviewed the four industrial revolutions, including the development of new technologies with time, adoption of the fourth industrial revolution by five leading goods manufacturing leaders (Germany, U. S. A., Japan, China, and Taiwan), their distinct strategies for smart manufacturing, and the technologies associated with smart manufacturing systems (such as virtual reality, augmented reality, mixed reality, additive manufacturing, big data analytics, Industrial Internet of Things, and artificial intelligence). Next, we discussed the influence of artificial intelligence on six aspects of the production line with practical case studies: quality checks, maintenance, reliable design, reduced environmental impact, factory integration, and post-production support. Furthermore, we also discussed inspection methods (such as conventional as well as artificial intelligence-based automated optical inspection) and the challenges in implementing smart manufacturing systems before future prospects in the global market are projected.

1. Introduction

Manufacturing industries are one of the key sectors with a major influence on the economy and growth of a country. Therefore, new technologies are continuously being developed to modify manufacturing processes and improve product yield and quality. A new technology has emerged recently in the manufacturing industries, popularly known as "smart manufacturing" [1], which can accurately predict product requirements and quickly identify errors, thus improving manufacturing processes and ultimately innovating products and services. Currently, this technology is playing a crucial role in producing better output with less human intervention while increasing the quality and sustainability of manufacturing activities while reducing costs. The smart manufacturing process involves the use of the Internet of Things (IoT), which establishes cooperative communication between machines and products and successfully incorporates customized instructions directly from customers [2]. This system is called cyber-physical system (CPS), which monitors the manufacturing process using computer-based algorithms based on a large number of combinations and coordination between physical and computational elements [3]. This technology uses smart devices and sensors for manufacturing goods, which in turn is highly beneficial due to significant reductions in the cost of the sensors, advances in sensing technology, and the utilization of highly efficient analytical applications that investigate the data for further decision making. This utilization of sensors in industrial applications is termed the Industrial Internet of Things (IIoT) [4]. Overall, smart manufacturing is full-integration and execution of all the aspects of manufacturing processes through exploiting data-based understanding, reasoning, planning powered by advanced sensing, modeling, simulation, and analytics technologies, which react in real-time in order to achieve the ever changing demands and conditions in the factory, in the supply network, and in customer needs [5]. The ultimate objective is to meet the needs for customized, smart, and environmentally friendly goods and services [6].

The current industrial revolution is called the fourth industrial revolution or Industry 4.0, which utilizes CPS to automate manufacturing processes and share information. Technologies such as IIoT, IoT, machine learning, artificial intelligence, and big data analytics have significantly improved data mining and industrial control. These manufacturing plants are also termed as "smart factories" [7]. The

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utilization of machine learning and cognitive computing in smart factories governs the real-time interaction between machines, people, and the entire production system. A large amount of data is collected from the production process, product defects, customer reviews, market demands, and analytics to make better decisions in order to optimize production and yield. Moreover, cloud computing allows networked data storage and data management, which can be utilized for off-site analysis, and the results can be accessed by the users through various mobile devices [8].

The successful operation of smart factories can be achieved by following three basic objectives: (i) the collection of a wide range of relevant data from the equipments, (ii) analysis of the data to generate useful information that can be used for operational as well as business decision-making, and (iii) acting on process automation, system optimization, and informing business strategy [9]. The components of a smart factory (Fig. 1) can be categorized into (i) smart production, which deals with the effective association of tools, machines, and operators; (ii) smart services, which include real-time monitoring of production/maintenance and predictive analysis; and (iii) smart machines, which deal with increasingly performing systems, manufacturing execution systems (MES) or shop floors, and material requirements planning (MRP) with waste reduction [10].

According to the McKinsey Global Institute's report, about 60 % of all occupations have at least 30 % of constituent activities that could be automated, which indicates that the effective utilization of smart manufacturing technologies can boost the production capacities of smart factories [11]. Smart manufacturing can be enabled in food, pharmaceutical, semiconductor, oil and gas, fine/specialty chemicals, paints and coatings, additives, etc. [12]. The global pandemic of Covid-19 since 2020 has impacted the manufacturing sector the hardest, making most manufacturers adopt smart manufacturing. With the rise in demand for day-to-day life goods along with electronic products, the global smart manufacturing market is expected to see a significant growth in the coming years. The global smart manufacturing market is projected to grow from US$ 249.46 billion in 2021 to US$ 576.21 billion in 2028, with a CAGR of 12.7 % between 2021 and 2028 [13]. Investment in automation technologies has increased significantly, which will further push the smart manufacturing sector.

2. Evolution of smart manufacturing

The manufacturing process has evolved significantly since the steam- and water-powered manufacturing processes in the 18th century to today's ultramodern digitalized manufacturing technology. The current industrial revolution is known as the fourth-generation industrial revolution, followed by a wide range of inventions and repetitive trials and errors over the years. The timeline for all four industrial revolutions is shown in Fig. 2.

2.1. First industrial revolution

Thomas Newcomen developed a prototype modern steam engine and revolutionized the industrial manufacturing process in the 1700 s. Following a few other innovations, the United Kingdom revolutionized the manufacturing process in the 1760 s, utilizing steam- or water-powered machines for the first time for the mass production of goods. Subsequently, between 1760 and 1820 or 1840, Europe and America as a whole also adopted these manufacturing processes [14]. Textile industries first implemented the most modern manufacturing techniques and attracted huge sum of investments [15]. Other sectors such as the iron industry, agriculture, and mining were also able to develop during this period. The main developments in the first industrial revolution were the invention of coal and iron, the formation of railroads, and the mass production of textiles. Such mechanization had a huge impact on the economic structure of society during that time. A highly efficient energy source was generated owing to the extensive coal extraction and the invention of the steam engine. The further development of railroads and significant growth in financial investments have accelerated the manufacturing processes. Despite having some advantages, such as advancements in innovations and inventions and related research, growth in goods production, construction of new transportation networks, etc., the first industrial revolution still had many drawbacks, such as deplorable labor working conditions, child labor, environmental pollution, and unhygienic living conditions [16].

2.2. Second industrial revolution

Some more technical innovations led the existing manufacturing processes to a new level, which is known as the second industrial revolution, occurred between 1870 and 1914, the beginning of World War

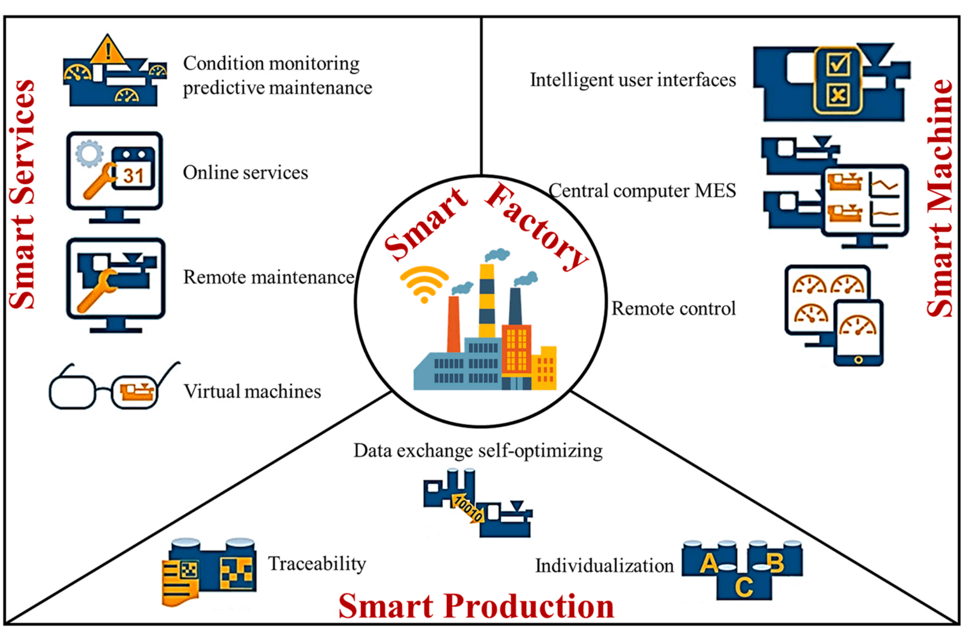

Fig. 1. Schematic diagram of the components of a smart factory [10].

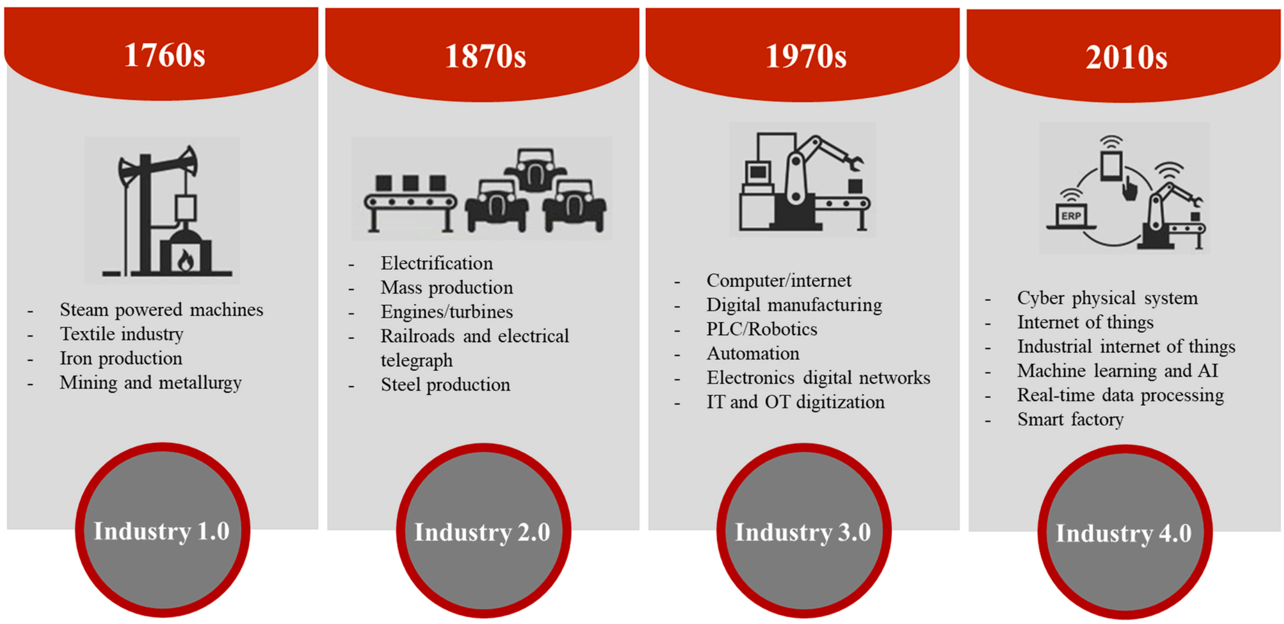

Fig. 2. Timeline and characteristics of different industrial revolutions.

I. The use of some new sources of energy, such as electricity, gas, and oil, reached its full potential via the development of new combustion engines. The massive development of railroads and electrical telegraph lines after 1870 led to exceptional growth in the sharing of ideas, transportation of manpower, goods, and manufacturing equipment, which led to a new wave of globalization, which in turn eased the manufacturing process and significantly increased the production yield. Later, significant innovations in electricity and telephones powered the manufacturing sector to a highly advanced level due to factory electrification and the production line. Electrical power expanded into a very common system of energy transmission, which led to the development of vacuum pumps, electric generators, gas lighting systems, transformers, etc. The steel industry also began to grow massively owing to the exponential demand for steel. New chemical synthesis techniques have also brought synthetic fabric, dyes, and fertilizer into the picture. Transportation methods were also revolutionized with the arrival of automobiles and commercial planes at the beginning of the 20th century [15].

2.3. *Third industrial revolution*

After the extensive effects of World War I and II, a significant reformation was required in the manufacturing sectors, which occurred after 1969 via the digital revolution. With the adoption of digital computers and digital record-keeping, mechanical and analog electronic technologies have rapidly been transformed into digital electronics. A new type of energy source, nuclear energy, emerged, whose potential surpassed that of its predecessors. This phase is known as the third industrial revolution or digital age. This also marks the beginning of the information age, which witnessed the development of microprocessors and transistors, as well as the massive use of telecommunications and computers. The innovation of new technologies has led to the development of miniaturized materials that have revolutionized space and biomedical research. The main feature of this phase is automation in production via the use of industrial robots and programmable logic controllers (PLC) [17]. The invention of the Internet and the World Wide Web further fueled the manufacturing process.

2.4. *Fourth Industrial Revolution or Industry 4.0*

Germany is the pioneer of the Fourth Industrial Revolution, which is also termed Industry 4.0 or I4.0. The journey began in 2011 with the promotion of the computerization of the manufacturing process. Cyber-physical systems (CPS) form the basis of Industry 4.0, which focuses on IoT interconnectivity, machine learning, and real-time data processing. The machines interact by merging IoT and industrial internet into the manufacturing system in order to share real-time information. The associated system algorithms help to make intelligent decisions during the manufacturing process. Artificial intelligence, augmented reality, flexible manufacturing automation systems, automated robots, and additive manufacturing are being widely and significantly adopted in the fourth industrial revolution [15]. I4.0 is characterized by (i) full automation in the manufacturing process, (ii) establishing IIoT- and CPS-based connections between the digital and physical world, (iii) optimized production steps based on customer demands, (iv) closed-loop data models and control systems, and (v) ordering products with customized features [18]. Enabling the autonomous decision-making manufacturing sector is the main attribute of the Fourth Industrial Revolution. Existing industries are gradually adopting such technologies to upgrade their manufacturing process into the idea of Industry 4.0.

3. Current state-of-the-art of smart manufacturing

After Germany in the West Europe introduced the concept of Industry 4.0 in in 2011, many other regions have also planned strategies to adopt smart manufacturing in their industry sectors. Following Germany's idea of the CPS-powered manufacturing process, as mentioned in Section 2.4, leading goods manufacturers, including the USA from North America and Japan, China, and Taiwan from East Asia have proactively begun to focus on new technological innovations for the same, which are summarized in Table 1.

3.1. *The United States*

General Electric (GE) introduced the concept of IIoT in 2012, suggesting that the future of manufacturing lies in implementing smart equipment, smart production systems, and smart and effective decision-

Table 1
Summary of the smart manufacturing strategies adopted by various regions.

Smart manufacturing strategies adopted by various regions			
Region		Strategy	Focus
West Europe	Germany	Industry 4.0[35]	Computerization of the whole manufacturing process via incorporating cyber-physical systems into the manufacturing system in order to share real-time data manufacturing process or product information.
North America	USA	Industrial Internet Consortium[36]	To combine OT which includes efficiency, utilization, safety, consistency, and continuity and IT which includes business insights, cost reduction, security, and flexibility, agility, & speed.
East Asia	Japan	Industrial Value Chain Initiative[37]	To increase collaboration between large manufacturers, small and midsized manufacturers, supporting member organizations, sponsor member organizations and academic members.
	China	Made in China 2025[38]	10 key sectors: (i) new information technology, (ii) numerical control tools and robotics, (iii) aerospace equipment, (iv) ocean engineering equipment and high-tech ships, (v) railway equipment, (vi) energy saving and new energy vehicles, (vii) power equipment, (viii) new materials, (ix) biological medicine and medical devices, and (x) agricultural machinery
	Taiwan	Productivity 4.0[39,40]	- Connecting to the local: fully utilizing the resources in Taiwan. - Connecting to the future: adding value to products, especially new technologies. - Connecting to the world: cooperating with other countries and introducing foreign professionals to Taiwan.

making [19]. A dedicated organization, the Industrial Internet Consortium (IIC), was formed in 2014 to accelerate the implementation of interconnected smart machines and intelligent data analytics [20]. The IIC was molded with the support of GE, Cisco, AT&T, Intel, and IBM with the aim of providing ideas, resources, pilot projects, and relevant activities about IIoT technologies as well as their security. The USA provides information technologies (IT), such as virtual reality, cloud computing, and big data, which are the most important for IIoT implementation [21]. Recently, IIC and CESMII, the United States' non-profit smart manufacturing institute for digital transformation, have collaborated to bring transformative business value for manufacturers via the development, adoption, and monetization of IIoT technologies [22]. The smart manufacturing innovation platform, profile, and marketplace brought in by these organizations would also accelerate the adoption of smart manufacturing by small and medium-sized enterprises. IIC will make the companies realize (i) increased shop floor visibility, which will identify expected issues even before they occur, (ii) intelligent supply chain management for inventory monitoring and process automation, (iii) decreased total cost of ownership by using smart grid technology in order to predict and optimize power consumption, (iv) streamlined human resources, resulting in fewer field service calls, remote process monitoring and diagnostics, and proactive equipment maintenance, (v) reduced environmental impact by improving energy efficiencies and material inputs, and (vi) increased profitability, leading to new business opportunities, more investments, and cost savings [9].

3.2. Japan

In response to Germany's Industry 4.0, Japan initiated its Industrial Value Chain Initiative (IVI) in 2015 [23] in order to design a new society in which manufacturing and IT are integrated. The top industries of Japan took part in this initiative. The purpose of the forum is to distinguish between collaborative areas (companies with common working strategies) and competitive areas (companies with competing technologies), organizing them, and interconnecting the unique technologies of the participating companies. As a result, manufacturing technologies can be connected irrespective of the manufacturing sites, companies, and countries, which in turn increases the values of the enterprises [24]. The IVI aims to identify the probable collaborative areas of participating companies and build a mutually interconnected system. The IVI carries out two types of activities: business scenario groups and platform workgroups. Some of the most important strategies of the business scenario groups are the utilization of robot program assets by CPS, traceability of quality data, collaboration among companies through shared process information, sharing technical information for horizontal integration of small and medium-sized enterprises (SMEs), and the predictive maintenance of presses and panel transportation devices. Platform workgroups are divided into several subgroups based on their intended work such as (i) production planning and control platforms, (ii) production engineering information platforms, (iii) quality management information platforms, (iv) supply chain management platforms, (v) small-sized enterprise information platforms, (vi) preventive maintenance platforms, (vii) asset and equipment management platforms, and (viii) maintenance service management platforms [23]. The IIC and IVI have signed a memorandum of understanding (MoU) to cooperate on measures to maximize interoperability between the production equipment and to secure the industrial Internet [25]. In addition to IVI, Japan has initiated another program called "Industry 4.1 J," where the "4.1″ indicates that the security level is higher than that of Industry 4.0. The strategies for the higher security level are (i) utilization of the control system to collect relevant data, (ii) record and accumulate data on the cloud platform, and (iii) ensure real-time analysis of the data by experts [26].

3.3. China

In 2015, China launched a 10-year program to achieve true technological independence and self-reliance as well as to enhance its manufacturing capabilities. The initiative is also known as Made in China 2025 (MIC25) [8,20,27]. The initiative aims to (i) improve innovativeness in the national manufacturing, (ii) merge industrialization and the relevant information, (iii) improve key industrial capacity, (iv) enhance the brand-building of Chinese goods quality, (v) encourage environmentally viable manufacturing, (vi) promote innovations in key sectors, (vii) ensure necessary restructuring of the already existing manufacturing sectors, (viii) establish high-end service-oriented manufacturing and related service industries; and (ix) encourage the involvement of international enterprises in manufacturing [27]. In the first step of MIC25, cloud manufacturing was proposed in China. As the basis of smart manufacturing, China has already established its system in certain areas via advanced computerized numerical control (CNC) machines, intelligent instruments, industrial robots, and additive manufacturing [21]. To encourage smart manufacturing applications, under the Ministry of Industry and Information Technology (MIIT), China funded 46 smart manufacturing based pilot projects in 2015, and 64 projects in 2016. These projects fall into five main categories: (i) discrete manufacturing, (ii) process manufacturing, (iii) network-based collaborative manufacturing, (iv) mass-customized manufacturing, and (v) remote operations and maintenance. A few Chinese companies have already implemented smart manufacturing in their factories-for example, Haier, a leading manufacturer of household electrical appliances, has adopted the concept of "connected factories," an ecosystem where the customers, products, machines, and factories are interconnected. Thus far, Haier has built 8 "connected factories" that have greatly reduced the time for product delivery from 21 days to just 10 days, with a 20 % reduction in operational costs [28].

3.4. Taiwan

Taiwan, which is known as the global hub for semiconductor chip manufacturing, has also embraced Industry 4.0, via various initiatives for smart manufacturing. Taiwan initiated the "Productivity 4.0″ plan in 2015 to push their industrial technologies so as to make Taiwan a significant contributor to the global manufacturing supply chain. The plan relies on six major strategies: (i) enhancing the smart-supply chain ecosystems of the leading manufacturers, (ii) providing necessary support for potential startups, (iii) rightly channeling production and services, (iv) realizing automated core technologies, (v) promoting talents with strong practical and technical knowledge, and (vi) introducing intelligent industrial policy tools [29]. Moreover, Taiwan is also home to most of the world's cutting-edge smart machinery, which is being developed under the "five plus two" innovative industries plan, introduced in 2016 by the government [30] to utilize Industry 4.0 technologies such as cloud computing, the Internet of Things, big data, and smart robots [31]. Taiwan, being the fourth largest machine tools exporter in the world, has a great potential for establishing smart factories. For example, Taiwan's Advantech's IIoT solution is widely acclaimed for its four-phase strategy of machine automation, process visualization, equipment networking, and predictive maintenance to help the manufacturer to adopt Industry 4.0 [32]. Taiwan's highly focused IT sectors and advanced manufactured goods reflect Taiwan's high degree of hardware/software integration [33], which also indicates Taiwan's potential to run smart factories. A dedicated research center for artificial intelligence for intelligent manufacturing systems (AIMS) was established by the Ministry of Science and Technology (MOST) of Taiwan in 2018 to develop next-generation intelligent manufacturing systems and push the development of smart factories in Taiwan [34].

4. Technologies associated with smart manufacturing

Many technologies are involved in efficiently coordinating smart manufacturing processes. The most significant ones are discussed below and are summarized in Fig. 3.

4.1. Immersive technologies

New techniques of generating, showing, and interacting with apps, content, and experiences are referred as immersive technology. This technology has revolutionized the digital experience by combining the senses of sight, sound, and touch. Currently, there are three kinds of immersive technologies available in the market which are virtual reality, augmented reality and mixed reality. In this subsection, we separately explain their contributions in the smart manufacturing.

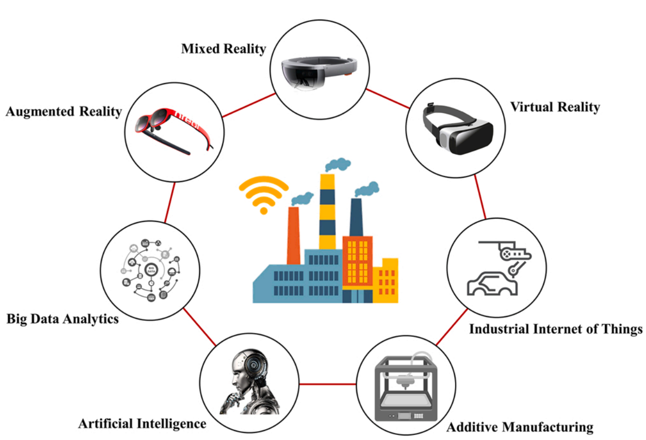

Fig. 3. Technologies involved in smart manufacturing.

4.1.1. Virtual reality

Virtual reality (VR) has seen noticeable adoption in manufacturing industries. VR technology merges computer-generated and simulated videos into real-world activities. It helps to visualize the product design from the initial stage without its physical presence, which reduces the prototyping time and costs. It also helps to digitally visualize the product's applications in a simulated target environment, which in turn broadens the options for product customization and other relevant modifications in the product design [41]. VR technology is also being applied in smart factories in automated guided vehicles (AGVs) and factories of the future (FoF) initiatives [42]. For example, Giorgini et al. demonstrated 3D visualization via a VR system of AGV movement on a target path in a simulated factory model [43]. Allmacher et al. presented a VR-based setup that utilizes motion capturing to sense and interact with humans via a smartwatch, which enables effective human-robot collaboration (HRC) via virtual commissioning [44]. However, such approaches are yet to be widely utilized in the manufacturing sector. VR technology also helps to train young engineers regarding industrial processes in detail, such as manufacturing processes, mechanization processes, and maintenance systems [45,46]. Thus, VR can significantly contribute to manufacturing education by realizing Internet of Things and Cloud technology.

4.1.2. Augmented reality

Augmented Reality (AR) integrates the real world with the digital world through wearables and mobile devices, where a computer simulation is utilized to create an artificial environment that digitally sees and analyzes any object and its usage. AR acts as a universal interface that improves human-machine interactions [47]. Various relevant videos, images, animation, sound, text, etc., are being combined in AR devices in a responsive manner [48]. Through the AR device, the operators can evaluate the ambient intelligence and react appropriately to the manufacturing related works in real-time. The obtained information can consequently improve the decision-making efficiency and quality, resulting in a better overall outcome [47]. It can also be used to analyze logistics performance, plan workshops, analyze products by observing their digital prototypes, and provide virtual training. AR does not require highly advanced expensive hardware and can be implemented using conventional equipment [49]. AR technology can also be used like VR to train new trainees in a more efficient way. By visualizing through an AR, a technician can easily get a clear idea of how to disassemble equipment, see the inner structures of the components, and perform servicing or repairs without prior experience [48]. Incorporating AR in a manufacturing process improves overall productivity. For example, warehouse workers in GE Healthcare use an industrial AR tool to kit and pick list orders up to 46 % faster. The AR tool is connected to a smart warehouse network to find the real-time location of the items, and the workers get easy-to-read instructions to locate them anywhere in the warehouse [50]. Lockheed Martin used the Microsoft AR tool to observe holographic representation of some complex components of aircraft and the necessary instructions to assemble them, which led to a 30 % reduction in the time required for component assembly by 30 % [50].

Both VR and AR technologies can help engineers to (i) accelerate the support in the production lines, (ii) help shop floor operators attain better equipment operation, (iii) provide instruction for assembling components via virtual training, (iv) monitor warehouse activities efficiently, and (v) support advanced diagnostics integrated into the modules [51].

4.1.3. Mixed reality

Mixed reality (MR) is where real and virtual worlds merge together and produce a whole new environment in which digital and physical objects co-exist and interact in real-time. Like the VR and AR counterparts, MR is also emerging in the smart manufacturing system. MR devices is highly capable to train the maintenance staff on the machinery they will be working with. In the virtual world, they can fully operate the machineries during training without actually operating them in the physical world. Which makes them a highly skilled personnel prior to applying their skills to the real-world machine. Moreover, maintenance staff can interact with the machinery in a MR world where they have schematic of the whole machinery which helps them to easily do modification of any faulty parts [52]. MR also presents an immersive visual representation of integration of machines, systems, and data that drives smart manufacturing. MR experience also enhances the product sales via virtual demonstration of product, work-instructions, service and maintenance and enables seamless knowledge transfer to the end users [53].

For better understanding of these immersive technologies, they are compared in Table 2.

4.2. Additive manufacturing

Additive manufacturing (AM), which is basically a computer-controlled 3D printing of products, plays a crucial role in smart manufacturing. Manufacturing plants have limitations in sample fabrication for research and development (R&D). The 3D prototype manufacturing of these products takes much less time with an accurate composition of constituent materials as well as the design parameters [54]. AM is highly beneficial for the rapid and accurate fabrication of complex components for research purposes. This accelerates the design process and mass customization, or the necessary changes in the subsequent steps [55]. This helps to optimize a product in terms of the target design, materials, related characteristics, and safety precautions according to market demand. The AM process can be remotely controlled online, where many 3D printing machines can be integrated together in the factory, and products, especially specified by users, can be produced within each machine [56]. Additive manufacturing saves a significant amount of time and cost during R&D. 3D manufacturing can also create the target application area of a product and help understand its overall performance. AM also helps smaller companies and end-users to design and develop their own innovative designs and prototype products by themselves as self-designers and manufacturers [57]. End users can order their desired design and customize the constituent materials, colors, and other aesthetic features online. They can also check the availability of materials and machines simultaneously. Hence, customers can easily upload their designs and obtain specific products [58]. Materials such as commodity and engineering polymers, metals, and their alloys, and sometimes metal oxides and ceramics are used in AM processes [59].

Table 2
Comparison between the three immersive technologies.

	Virtual reality	Augmented reality	Mixed reality
Visual type	Fully artificial environment being disconnected from physical world.	Virtual information is projected onto physical environment.	Virtual environment is merged with physical environment
Interaction type	Interaction happens in virtual world.	Interaction happens in physical world.	Interaction happens both in virtual and physical world.
Setup	Headsets are required.	Headset, smartphone, head-up display are required.	Gears used in VR and AR are required.
Pros	1. Great tool for virtual communication with people. 2. Provides very detailed 3D view of the virtual world. 3. Very effective in learning the detailed parts of the machineries.	1. Users can physically see the real world. Therefore, the users can easily move around both indoor and outdoor when wearing the gears. 2. Training in AR is highly effective due to it merges with the real world. 3. It prevents expensive process reworking.	1. Users are still in touch with real world. Therefore, the users can move around in a limited space, mostly indoor. 2. Provides merits from both VR and AR for learning and training new process and machineries. 3. It drastically reliefs the loading of equipment maintenance.
Cons	1. Totally disconnects from the real world. Therefore the user is not supposed to move around when wearing the headsets. 2. Training in VR environment is not real.	1. Headsets do not have very high brightness compared to the ambient environment. Therefore the visuals are not distinct at all time. 2. Narrow field of view.	1. Very expensive technology. Difficult for SMEs to afford. 2. Efforts required for preparing activities in the real world.

4.3. Big data analytics

The smart manufacturing network is highly complex, which in turn generates new challenges. This opens a window for innovation research and development activities. Big data analytics is a technique that significantly influences smart manufacturing via the collection and analysis of a large amount of data from the production units, customer feedback, and product request system, which helps in real-time decision making [14]. This analysis of big data (i) generates knowledge in continuously changing data, (ii) enables the observation of value streams, which helps to monitor and detect any irregularities in the values, (iii) increases the effectiveness of model optimization, which acts as the basis for prediction of happenings, (iv) enables new kinds of diagnosis possibilities, and (v) helps to optimize KPI [60]. Big data is being utilized by a wide range of manufacturers. For example, General Electric's New York factory is linked to other factories worldwide via advanced sensors to collect data on manufacturing processes and building conditions. Employees can assess these data in real-time and receive alerts regarding production difficulties or anomalies. This helps to rapidly improve the quality and efficiency of the overall process. Intel successfully reduced the amount of quality testing by using big data in predictive analytics, ultimately lowering the cost of chip and processor manufacturing. BMW can identify design weaknesses by collecting data from sensors on prototypes during testing [61]. Thus, big data analysis will benefit manufacturers by analyzing and predicting failures in real-time.

4.4. Industrial Internet of Things (IIoT)

The Industrial Internet of Things (IIOT) is highly essential in the evolving Industry 4.0. In this technology, the entire process of a production line is attached to sensors to monitor the equipment performance as well as the process output. The sensors are connected via the Internet cloud to enable the integration of all the process steps [56]. This integration helps in better production planning via smart process control of tools and materials optimization, predictive maintenance, discovery of machine faults, and better human-machine interaction [62]. IIoT enables interconnected digital supply networks (DSNs), which transform normal factories into smart ones by making them highly efficient, saving energy and costs, and reducing operational risks [63]. IIoT utilizes machine learning and big data analytics to extract and analyze the data from all the sensors, enabling factories to detect various complications and problems, and thereby saving production costs and time. IIoT acts as a core technology in smart factories and is highly beneficial for quality control, equipment fault prediction, supply chain traceability, and supply chain efficiency [4]. The extracted data from the sensors are stored in the cloud via IoT connectivity, and further analyzed and shared with authorized stakeholders after combining with the necessary information. IIoT significantly improves manufacturing outcomes by increasing the yield and reducing material waste, hence speeding the production while maintaining the high quality of the goods.

4.5. Artificial intelligence

Artificial intelligence (AI) has been adopted to achieve better utilization of robots and significantly reduce human involvement in risk zones. AI also helps to better maintain the production line and detect machinery or product failures. AI-powered systems are capable of self-decision making, self-optimization, responding automatically to changes in production schedule and malfunctions of any machines, automatically replacing any machine components, and transmitting alarms for any uncontrollable situations [64]. AI technology enables a high degree of automation, mass customization, employment of instinctive robots, and intelligent machines, etc. [65]. Performance enhancement of the entire factory, product cost control, process optimization, and improved efficiency can be realized using AI technology [66]. Using predictive and prescriptive analytics through machine learning effectively improves assembly workers' lives by bringing predictability and certainty to operations [67]. With human–robot collaboration, AI can learn the predictive tendencies of the facilities and solve complex problems. For example, when it is necessary to precisely control temperatures, pressures, and liquid flow, humans are very prone to making mistakes. However, AI-based systems can easily control operational processes and make necessary decisions on critical factors such as security, safety, and productivity efficiency [68]. Downtime in a continuous production line is highly detrimental to the final product yield, which can be avoided by utilizing AI-powered systems [67]. According to Infosys, most companies want to automate manufacturing via AI-based systems to enhance production yield, considerably reduce manual errors, and reduce people's involvement in repetitive tasks. The rapid development of AI has also brought digital twin technology to the manufacturing sector, which was designed to increase the performance of physical things through the use of computational techniques offered by virtual replication [65,69]. If design flaws are discovered while utilizing the traditional irreversible design approach, digital twins can realize hardware-in-the-loop simulation of both cyber models and physical equipments in order to avoid the significant cost of manufacturing system reconfiguration. A CMCO (i.e., Configuration design-Motion planning-Control development-Optimization decoupling) design framework has been followed for designing the flow-type smart manufacturing system. Configuration model deals with the topological structure and static configuration of the flow-type smart manufacturing system. Motion model refers to motion of the equipment and movement of the work-in-progress. Control model refers to the autonomous control system of the flow-type smart manufacturing system. Optimization model is the optimization of the problems and their coupling in the manufacturing process. The digital twin based CMCO architecture can realize hardware-in-the-loop simulation which can avoid the possible design errors in the very early stage of the manufacturing system. Integration of quick-response or artificial intelligence algorithms to perform the dynamic optimization can improve the system operation efficiency by analyzing the online available data. [70]. Industry 4.0 is powered by an AI system that utilizes highly advanced deep learning for automatic processing of data, speech recognition, image reconditioning, natural language processing, multimodal image-text, and games. It automatically processes the highly complex and nonlinear data via cascade of multiple layers. Deep learning controlled manufacturing system is enabled to learn automatically from numerous data and identify patterns, and make decisions by mining knowledge. Data analytics such as descriptive analytics, diagnostic analytics, predictive analytics, and prescriptive analytics can be realized. Descriptive analytics summarizes the product's conditions and operational parameters. Diagnostic analytics examines the root cause of the product's degraded performance or the equipment failure and report the reasons behind. Predictive analytics explores various statistical models based on historical data to predict the possible future degradation in production or equipment. Prescriptive analytics recommends the employable one or more plan of action. These advanced analytics transforms manufacturing process into highly optimized smart facilities which makes deep-learning a powerful analytic tool for smart manufacturing system. In the era of big data, it boosts data-driven manufacturing technologies [71–73]. In addition, machine learning (ML) plays a key role in Industry 4.0 for recognition of patterns, detection of faults, quality control, and monitoring. In ML, all the data is divided into training and testing data set. The ML Model is trained using training data, which covers 70–80 % of the data, and then, the overall performance is evaluated based on rest if the 20–30 % test data. The models are utilized to handle high-dimensional, complex, structured or unstructured data in order to find correlation between the large data sets. ML algorithms have three categories including (i) supervised learning, (ii) unsupervised learning and (iii) reinforced learning which perform two main task, classification and regression. ML is very well explored in predicting atomic arrangement in the materials which helps in further designing the materials by considering chemical composition, process parameters and microstructure. Various machining processes, especially in milling and turning, ML algorithms have been used very frequently. Amongst all, artificial neural network (ANN) is most commonly used in cutting process [74]. Other algorithms such as support vector regression (SVR) and random forest (RF) are also used. Utilization of ML models in machining process significantly improves the productivity and product quality. During quality monitoring, ML models such as Naive Bayes and decision tree classifier can classify good and defective products by analyzing the images or surface roughness data. Predictive maintenance is carried out using ML in supervised approach, where it helps to predict the failure of tools or equipments before it happens. If the past failure data is available, if it is continuous then regression problem can be implemented and if the data is in the form of categorical value then classification method can be applied. Machine learning can also be utilized for calculations of mechanical properties such as stress, strain, and the effect of temperature on the mechanical behavior using finite element analysis results [75]. As the industries are moving towards complete automation, the influence of data management and data processing has become very important and that is where the dedicated machine learning algorithms play significant roles [76].

5. Influence of AI in smart manufacturing

5.1. Quality checks

Quality inspections are the most important step in a manufacturing process because of their ability to sort out undesired products before they go to market. However, finding effective ways to perform accurate quality inspections is challenging. Manual inspections are time-consuming and can result in inaccurate and inconsistent product quality due to unavoidable human errors. Therefore, highly advanced quality checkups throughout the whole supply chain are very important for customer satisfaction as well as loyalty, as inspections are performed in the production line. Machine vision technology is an imaging-based inspection technology where machines, robots, and autonomous devices can see, detect, and analyze images automatically, and can automatically inspect product quality and provide data from the manufacturing floor to warehouse fulfillment to the distribution center. Machine vision integrates high-resolution smart cameras, artificial intelligence, and edge computing [77]. AI-based machine vision systems are able to detect defects and mistakes on packaging and labeling, and can also make necessary decisions based on the situation. Standard machine vision can detect if there is anything wrong; however, it can neither classify it nor take necessary action. AI-based machine vision becomes smarter as it collects more images and classifies the defect and takes necessary actions accordingly. Machine-vision-based automatic optical inspection (AOI) plays a crucial role in manufacturing and processing in industry, and it has been found that AOI not only improves product quality and productivity, but also increases competitiveness [78]. A typical AI-based machine vision setup is shown in Fig. 4, where a smart camera is equipped with all the necessary sensors and AI integrating software. The camera is also being connected to a cloud storage of similar images. The whole system prepares itself for the image capture. When required, the working distance is changed automatically by zooming in and out. This mechanism is called the pre-processing automation. As the image is captured it's being processed via the image processor. In this step the images are being prepared in a way such as converting color images to grayscale, standardizing images to have identical dimensions, and other transformations. Later various algorithms are utilized to analyze the extracted data from an image in order to create predictions. Many defective products will be rejected by the algorithm when image processing is done [79].

5.2. Maintenance

In the era of Industry 4.0, smart factory floors are equipped with sensors that collect huge amounts of data and capture a multitude of activities. The collected data are analyzed in real-time by analytic capabilities, which can eventually help in decision making and provide inputs to reduce untimely downtime in a production line, enhance manufacturing efficiency, automate production facilities, predict probable market demand, optimize inventory levels, and improve risk management. Prognostic and Health Management (PHM) systems are controlled by AI to detect whether any equipment has problems in usual operating conditions or when a fault occurs [80]. With the increasing availability of data over time and the recent progress in machine learning and deep learning, significant improvements have been found in the performances of AI-driven PHM systems. Such predictive maintenance is very helpful in evaluating the real-time condition of in-service machines to predict when maintenance is required. PHM utilizes data science, statistics, and physics to identify faults in the system, followed by classifying them according to their specific types (diagnostic). In the subsequent steps, PHM predicts and forecasts how long the system of the machines will be able to work in the presence of faults (prognostic) [81]. Such tools allow maintenance to be performed only when required, which can be predicted much earlier. This makes maintenance a cost-effective and less burdensome process. Predictive maintenance prevents unanticipated equipment failures [82]. The basic steps of the PHM system are shown in Fig. 5 [83]. The PHM system typically consists of three basic steps namely (i) observe, which involves data acquisition and processing, (ii) Analyze, which includes condition assessment, followed by diagnostic and prognostic analysis, and (iii) Act, which is done by human and machine interaction backed by all the decision support tools.

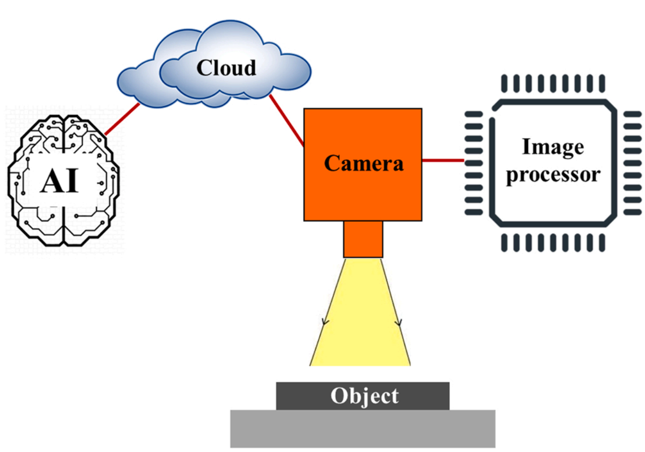

Fig. 4. A typical machine vision system for product quality check.

5.3. Reliable design

In modern smart manufacturing, AI is also being used to generate thousands of feasible designs of any entity by providing the previous design problems, basic dimensional parameters, maximum weight it must withstand, mechanical strength, and constituent material options. This is known as generative design, where AI-based software explores all possible design combinations from the input data. A few companies such as Airbus, Under Armour, and Stanley Black & Decker use such design technology to develop design ideas that are beyond the considerations of the human mind [84]. Such designs provide a significant number of design options that can reduce the amount of materials required and the product weight while maintaining their strength and cost [85]. Generic algorithms can be useful at all stages of end-to-end product development, such as testing new geometries and shapes for their viability in the final product with desired characteristics, achieving higher levels of product sustainability and performance while minimizing the cost, and checking the manufacturability of the potential designs [86]. The designs are made into product samples by additive manufacturing, which helps to fine-tune the design by refining and reprinting a product as many times as needed until ready for production [87]. This design process can also be integrated with IoT, which helps to consider customer feedback for future generative design strategies. Thus, generative designs can help to realize innovative, cost-effective, user-friendly, and sustainable products. A real-world design series developed utilizing AI is shown in Fig. 6 [88].

5.4. Reduced environmental impact

There is no doubt that the manufacturing sector contributes significantly to worldwide energy consumption, up to 50 % of the total [89]. Most of the waste is from poor-quality products as well as unused raw materials in the production line. According to the International Energy Agency, energy demands are increasing with time [90]. However, according to a report by Microsoft and PwC, AI can immensely benefit environmental sustainability and guide the manufacturing sector to become more eco-friendly and energy-efficient [91]. AI can solve issues such as reducing the excessive use of certain materials that are not good when disposed into the environment, optimizing the production of scrap waste, and evening out the unequal distribution of energy resources among the logistics [92]. The use of robots allows the work to be done precisely, which reduces the energy used. The integration of generative design and additive manufacturing helps to produce less waste and has fewer adverse effects on the environment than do traditional manufacturing processes for sample manufacturing [93]. Effectively used AI helps to determine product quality issues and to optimize products at inventory levels and the use of resources in the production line [89]. Thus, when manufacturers use AI to produce high-quality products, manufacturing waste will automatically be reduced.

5.5. Factory integration

In smart factories, everything from material selection to product delivery is being integrated via IIoT, IoT, and AI, and at the same time, a few enterprises are also integrating different manufacturing plants together in order to increase the overall yield. The data gathered from one production line can be analyzed and shared with other factories to automate material provision, maintenance, and other manual undertakings for great savings of time and energy [94]. These are enabled by the communication between machines and humans across all the factories. This system integrates customer demands, process design, and products, and enables continuous delivery of value to a company's customers [95]. Haier established such connected factory networks all

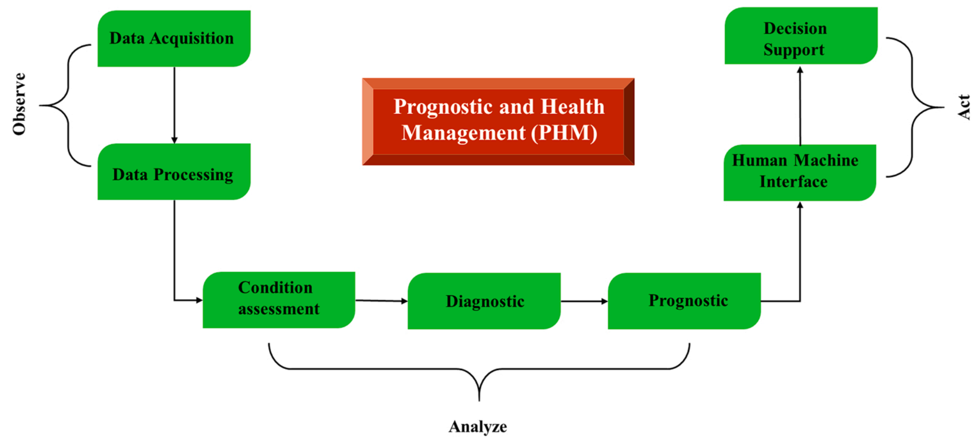

Fig. 5. Basic steps of a basic PHM system.

Fig. 6. AI-powered generative design models.

over China, which are based on vertical and horizontal dimensions. Vertically connected factories increase customer satisfaction by meeting specific demands. Horizontally connected factories require flexible manufacturing capabilities to realize customized products. This requires automation, modularization, and digitization of production lines and reliable business processes [28].

5.6. Post-production support

AI can also be used to update clients about product performance. KONE, an elevator and escalator manufacturer, uses an IoT platform and AI tool to monitor and predict the real-time condition of their elevators and escalators from the attached sensor data. This allows them not only to predict defects, but also to show clients how their products are being used in real life [94]. The data are processed and analyzed in a cloud platform to recognize equipment condition usage patterns and potential probable faults that could follow. The AI used with KONE 24/7 connected services can predict when a collapse is likely to occur and/or any servicing is needed [96].

6. Automated Optical Inspection (AOI) in smart manufacturing

In the Industry 4.0 era, highly accurate quality control is one of the most important segments in any manufacturing plant because inspecting the products before they go to market is an effective way of preventing supply chain disruptions and maintaining high quality standards. The inspections are mostly carried out via the AOI system because of its non-destructive technique (NDT). Human eyes are capable of sensing electromagnetic wavelengths ranging from 390 to 770 nm. However, video cameras can be customized to sense a much wider range of wavelengths [97].

6.1. Conventional AOI system

The AOI is an integral part of quality control in Industry 4.0. What makes the AOI system different from manual optical inspections is the recent rapid developments in image capturing and processing technologies that improve the precision of AOI-based quality inspection. A fully automated optical inspection system comprises both hardware and software setups. The hardware setup consists of image sensors and illumination settings that acquire digital images of the product. The software setup includes various inspection algorithms that analyze the images and notify defective and non-defective products [98]. Recently, various 3D imaging devices that use fringe projection have been introduced [99,100]. Recently, 3D-AOIs have been able to identify difficult soldering problems that are not so easy to detect, such as checking lifted leads of surface mounted devices (SMD) or coplanarity checking in ball grid arrays (BGAs), which are almost impossible for typical top-view cameras to detect. Moreover, owing to the advancements in Industry 4.0 technologies, new AOI systems not only produce but also store vast amounts of data, which improves the potential of inspection machines [101]. Currently, conventional AOI systems are mostly used to inspect printed circuit boards (PCBs) to detect nodules, scratches, stains, open circuits, and the thinning of soldered joints [102]. In addition to PCBs, AOIs are used for inspecting semiconductor wafers, light-emitting diodes (LEDs), and flat panel displays (FPDs; e.g., LCDs and OLEDs). The inspection items include surface level defects, image analysis, pattern matching, color verification, product dimensions, assembly verification, and optical character recognition/verification blob analysis [103]. Fig. 7 shows a schematic of the conventional AOI system.

6.2. AI-Powered AOI

With time, electrical/electronic device designs have become more complex, and for quality checks, conventional AOI systems are likely to make more mistakes during inspection. An AI-based AOI system learns from a vast amount of image data acquired from good and bad products. The AI software builds a model with a ruleset that allows the in-build algorithm to understand where and what specifically to look for in a very efficient way [104]. The more it scans products, the better its accuracy becomes. AI-based high-resolution cameras can perform quality checks at any point of the production line and find the defects in real-time, which helps to accurately identify the origin of defects, if any, in the early stages of manufacturing [105]. Therefore, many manufacturers have recently integrated machine learning techniques and deep learning with the AOI systems in their smart factory production lines, which significantly improves the result and speed of the detection process [106,107]. Thus, AI-powered AOI improves the yield and fabrication processes and significantly reduces manual operations [108]. Lo et al. suggested a process for an optical inspection method in which they determined the pattern transfer completeness (PTC) of inkjet-printed electrical devices, in which the line edge roughness (LER) of the printed patterns was calculated through image operations [109,110]. They also determined the effect of the LERs on the electrical properties of the devices and determined a relationship between pattern integrity and electrical characteristics using a machine learning model [111]. This model can be used as an AOI system for smart factory production lines. An AI-based AOI system supplier has claimed a quality inspection accuracy rate of 97 % compared to the 60–70 % accuracy of conventional AOI systems [105]. AI-based 3D-AOI considerably reduces the number of false alarms in solder defect inspection, which is very difficult to achieve with conventional 3D-AOI technology, significantly improving the accuracy of solder defect detection in a smart factory [112]. Fig. 8 shows a schematic diagram of the AI-powered AOI system.

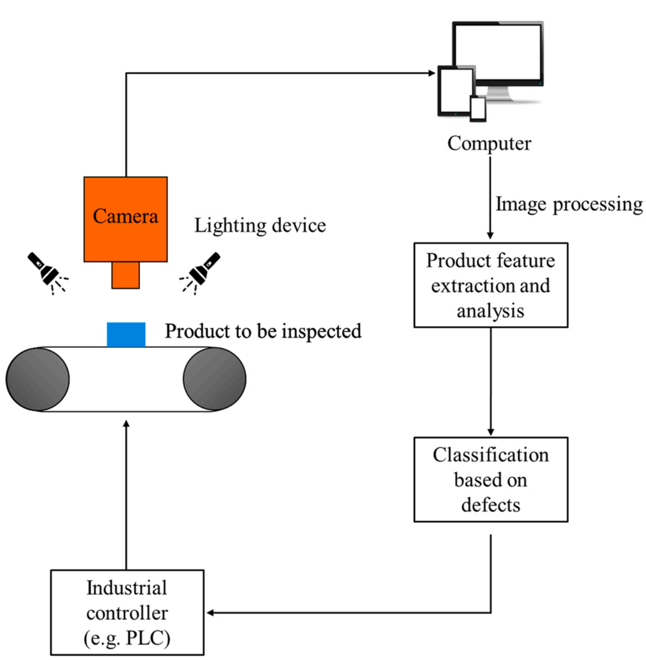

Fig. 7. Schematic diagram of a basic AOI system.

7. Challenges in smart manufacturing

Regardless of how advanced a system is, it always carries many serious security concerns. Even after using modern cutting-edge technologies in many factories to achieve smart manufacturing, there are many challenges faced by enterprises. There will always be new risks as smart manufacturing capabilities are based on levels of technical complexity, integration, and automation well beyond those of traditional manufacturing processes, and the lack of clarity on security is equally troubling. The rise in cyber-attacks in smart factories demonstrates that even current systems are susceptible to many security threats, which are poorly understood. Consequently, enterprises are not well prepared for the potential security dangers that exist [113]. Especially SMEs experience very hard time to deal with such obstacles due to their budget constraints [114].

7.1. Data security

Data security is crucial for smart factory adoption, where IIoT and IoT are extensively used to connect operational technology (OT), information technology (IT), and intellectual property (IP) [115]. The combination of physical and virtual systems in smart factories enables interoperability and real-time functioning. However, it comes with the risk of cyberattacks. There are always possibilities of data breaching when they are shared on the Internet. Misuse of these highly classified data can always pose serious threats to the manufacturing sector. One study demonstrated that 65 % of companies consider IoT technologies to mostly lead to cybersecurity risks [116]. Even a single cyberattack can completely collapse a smart factory and cut access to real-time data monitoring, predictive maintenance, and supply chain management. Therefore, impenetrable cybersecurity is in high demand in smart factories. In a report by Deloitte and the Manufacturers Alliance for Productivity and Innovation (MAPI) in 2019, they revealed various risks associated with smart factories, and found that 48 % of the surveyed factories found operational risks, including cybersecurity, as their greatest threat, one that can easily disrupt production, business, and the market [117,118]. Manufacturers should provide cybersecurity not only to smart factory operations, but also to all employees. Organizations should employ layered security approaches that can secure all the networks and quickly defend each component of IT and OT systems when attacked by a third party [119].

7.2. System Integration and Interoperability

Smart factories tend to keep updating their machineries with the newest technology equipments. However, existing machines may face compatibility issues with new ones. The most challenging task is to establish machine-to-machine (M2M) communication. For example, the current manufacturing systems perform the best under IPv6 connectivity and support devices connected at the same time, which might not

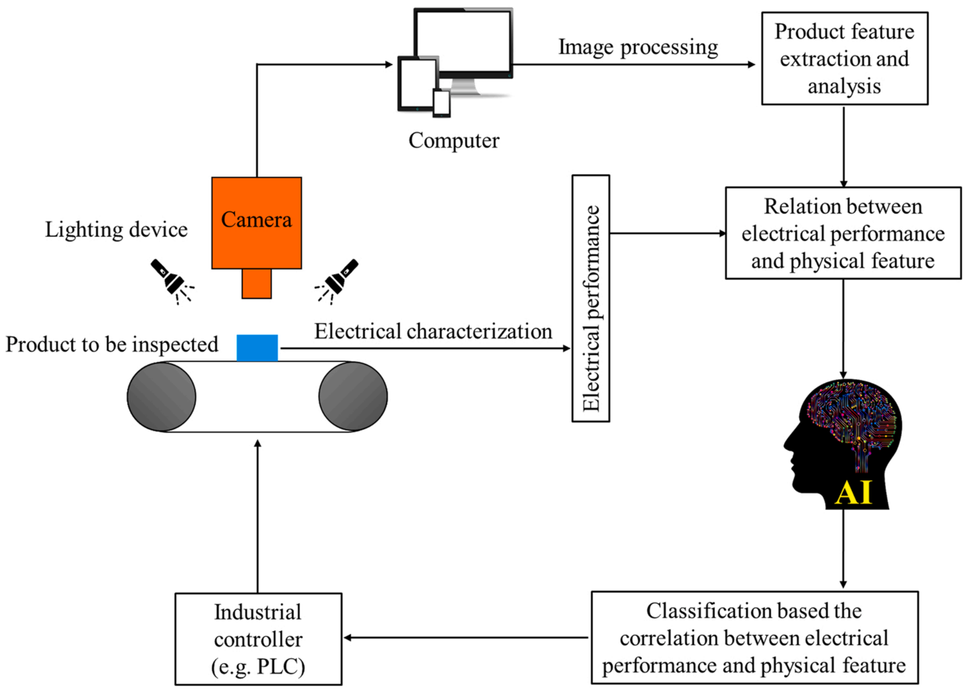

Fig. 8. Schematic diagram of an AI-powered AOI system.

support the previous edition of the equipment [14]. The major concerns of autonomous machine interactions in the context of smart manufacturing are ubiquitous M2M messaging and machine understanding [120]. In addition, a lack of clear understanding of the return on investment (ROI) of the new technologies prevents manufacturers from investing huge sums [121]. Moreover, the introduction of the latest AI and IIoT solutions does not necessarily add value to smart manufacturing. To build an effective smart factory, besides introducing new technology hardware and software, it will be even more significant if their practical experience in solution introduction is being analyzed [122]. The implementation of these technologies should also enhance the interoperability of systems. The effective interoperability between machines depends on the following factors: (i) compatibility and data transfer between the software, even made by the same vendor, but having different versions, (ii) misinterpretation of the terminologies used for data exchange between the machine, (iii) inconsistent data formats or standards, and (iv) connectivity in the IoT realm [123,124].

7.3. Safe human-robot collaboration

Collaborative robots (cobots) play an integral part in Industry 4.0, and significantly enhance the overall efficiencies of factories by drastically reducing human errors. Cobots can interact directly and safely with humans, even in a manufacturing environment, and potentially improve human-machine interactions [125]. These cobots are mostly programmed to perform specialized tasks in smart factories [126]. However, the safety of the associated personnel on-site can be a major concern in case of any failure of the cobots. The risk can also occur from the human side if the personnel are not trained properly for collaboration and work with the cobots. Moreover, mobile cobots are powered by batteries, which are likely to lose charge in the middle of a task, causing serious downtime for manufacturers [127]. The manufacturer should program the cobots so that they return to the charging station by themselves and connect when their charge falls below a predefined threshold. Relying on such collaborative robots can backfire, as they are not very efficient in differentiating between raw materials [128]. Cobots generally work under human supervision. They are not fully autonomous, and require checks to be performed at particular times. Therefore, trained personnel must be present with the cobots to provide necessary guides, which can be very tiring to employees [129].

A summary of the literatures has been presented in a tabulated form in the Table 3. The tables shows the objectives, question or problem raised and whether they were solved or not in those research works.

8. Future prospects of smart manufacturing in the global market

Since the COVID-19 pandemic, manufacturers have expressed high demand remote working and improvements in supply chain agility, which can be achieved by implementing smart manufacturing technologies [14]. According to a recent study in 2021 involving nearly 300 manufacturers, 83 % of them mentioned that smart manufacturing technologies and processes are their priority, and over 60 % of them plan to begin adopting smart manufacturing technologies within the next 12 months [130]. Manufacturers would prefer employees with the relevant technical knowledge and skills, who can easily adapt to new technologies to efficiently run smart factories. The most important skills are communication and teamwork, flexibility, analytical thinking, and knowledge of smart technologies. In the coming days, SMEs are highly like to adopting smart manufacturing technologies [131]. It will be much easier for them to adopt the smart manufacturing process if they follow a framework which includes, (i) manufacturing data identification available in the SME, (ii) smart manufacturing readiness assessment of the SME among the data-hierarchy steps, (iii) developing a smart manufacturing tailor-made objectives for the SME, and (iv) identification of the suitable tools and practices that will lead towards the tailor-made objective. SMEs adopting this framework will push them towards implementing a data-driven decision-making culture. [132] However, the major characteristics of smart factories, which are self-optimization, self-configuration, predictive maintenance, and decision making, are still lacking in many advanced factories [14]. 5 G technology and edge computing will surely boost cloud manufacturing and big data analysis. With the help of IoT, end users can directly request services according to their own demands [133]. Thus, networked manufacturing services enable smart decision making. Big data analytics will also accelerate the extraction of relevant knowledge and information from an extensive amount of production data and support specific

Table 3
Summary of literature review.

Literature reference number	When contributions have appeared in the literature and with which contents	Objectives posed by the authors	Questions or open problem raised	Problems solved or not
43	- Technologies Associated with Smart Manufacturing - Virtual reality	Visualization of automated guided vehicles (AGVs) in virtual reality and collision detection with large scale point clouds	Collision detection of AGVs in warehouses and real-time visualization in VR.	Yes, three types of collision were tested with different signals for each type. Each collision event triggered the vehicle to stop until a resume command is issued by the user.
44	- Technologies Associated with Smart Manufacturing - Virtual reality	Development of Virtual reality for virtual commissioning of automated guided vehicles	Problem in developing AGV software due to rising autonomy.	Yes, the work validates the human-robot collaboration (HRC) functionalities of an AGV via virtual commissioning.
68	- Technologies Associated with Smart - Artificial Intelligence	Designing of the digital twin-based configuration, motion, control, and optimization model of a flow-type smart manufacturing system	Difficulty in achieving the expected effect of the design and implementation of smart manufacturing system.	Yes, A quad-play CMCO architecture is put forward for the designing of the flow-type smart manufacturing system in Industry 4.0.
78	- Influence of AI in Smart Manufacturing - Quality Checks	Developing a machine vision based automatic optical inspection system for measuring drilling quality of printed circuit boards	Difficulty in inspecting holes on PCBs faster and efficiently.	Yes, the work have used linear motors, and a CCD camera to achieve an AOI system to detect the positions and defects of holes on the PCB
83	- Influence of AI in Smart Manufacturing - Maintenance	Enabling Health Monitoring Approach Based on Vibration Data for Accurate Prognostics.	To extract features that explicitly reflect failure progression or have essential characteristics such as monotonicity and trendability.	Yes, The proposed approach is generalized on the vibration data from two real applications, i.e., cutting tools and bearings. The results clearly depict the effectiveness of the proposition for improving the accuracy of prognostics.
88	- Influence of AI in Smart - Reliable Design	Innovative approach to the design of mechanical parts.	Mechanical parts designed in CAD programs are limited depending on the ability of the designer and the program used.	Yes, the results obtained, unlike traditional design, reveal many lighter and optimized design results compared to different materials and manufacturing processes.
101	- Automated Optical Inspection (AOI) in Smart Manufacturing - Conventional AOI System	Development of Intelligent Optical Inspections	Programming the inspection machines is time consuming and requires a good understanding of the optical inspection domain and printed circuit boards.	Yes, a novel and integrative approach for the intelligent optical inspection of PCBs has been proposed.
107	- Automated Optical Inspection (AOI) in Smart Manufacturing - AI-Powered AOI	Developing a rapid recognition method for electronic components based on the improved YOLO-V3 network	Balance between detection speed and accuracy in rapid object recognition was not well solved.	Yes, the experiment was completed, and the proposed method was compared with several popular detection methods. The results showed that the accuracy reached 95.21 %.
109	- Automated Optical Inspection (AOI) in Smart Manufacturing - AI-Powered AOI	Developing an advanced qualification method for patterns with irregular edges in printed electronics	Difficulty in accurate evaluation of pattern transfer completeness (PTC) due to irregular line edge roughness (LER).	Yes, the study proposed and examined an advanced method which quantifies and qualifies PTC in terms of the LER.
110	- Automated Optical Inspection (AOI) in Smart Manufacturing - AI-Powered AOI	Developing efficient and improved qualification method for patterns with irregular edges in printed electronics	Irregular edges in printed patterns not only lead to production challenges, but also raise uncertainties concerning the performances of electronic devices	Yes, the results show that the proposed method correctly evaluates the PTC of printed patterns with on average 97.6 % efficiency enhancement, with at most 37.7 %.
111	- Automated Optical Inspection (AOI) in Smart Manufacturing - AI-Powered AOI	Prediction off-line electrical characteristic through in-line pattern integrity inspection by utilizing machine learning-based model.	No relation had been established to predict the electrical performance of a microelectronic component from its specific pattern integrity.	Yes, the study predicts the off-line electrical characteristic (capacitance) of a microelectronic component (antenna in two-arm Archimedean spiral shape) from its in-line physical features based on a model built through machine learning.

decision-making [134]. Advanced technologies or algorithms, such as deep machine learning (DML) and AI, will play a significant role in the same [135].

Furthermore, in the age of global warming, manufacturers are committed to achieving carbon neutrality because being environmentally friendly is also one of the main attributes of smart factories. To reduce carbon emissions, companies should operate their factories with optimized energy consumption via highly efficient manufacturing processes by adopting computer simulation/modeling and smart management [136].

9. Concluding remarks

Smart manufacturing is a very important paradigm for the fourth industrial revolution, one that holds the promise of customization according to the end user, superior product quality, and improved factory productivity. In this article, we review the evolution of industrial manufacturing technologies and the current scenario of smart manufacturing, which is an integral part of Industry 4.0. We have also elaborated on various smart manufacturing technologies and emphasized the influence of AI technologies on the entire manufacturing process. Current challenges are also discussed, which can provide future research directions to academia and practitioners.

Manufacturers are opting for full automation of factories to reduce human labor and increase the pace of production. AI-powered quality inspection technologies also play a crucial role in allowing the best quality products in the market. Moreover, new concepts such as immersive technologies, additive manufacturing, IIoT/IoT, and CPS have played significant roles in uplifting manufacturing processes equipped with smart manufacturing systems. However, these technologies are not very common in most manufacturing sectors, leading to a large gap between the conceptual smart manufacturing system and the existing manufacturing system. This can be due to the difficulty and lack of knowledge in integrating smart manufacturing technologies as well as the return on investment in the same. Adoption of smart manufacturing can be challenging at the beginning, but once it has been introduced, it offers great profits to enterprises as well as the end users.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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